

P Akshay Thomas

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EDUCATION

Indian Institute of Technology Madras
B.Tech in Computer Science & Engineering

Jul 2018 - May 2022

EXPERIENCE

RINGCENTRAL INNOVATION INDIA

Sept 2022 – Present

Software Development Engineer 2 (Integrations & Automations Team)

- Architected a **Multi-Agent Platform** using **LangGraph** and **FastAPI**, with a **Dynamic Schema Registry** in **PostgreSQL** that decouples agent definitions from the codebase — enabling instant agent registration without redeployment and reducing onboarding time by **90%**; delivered real-time token streaming via **SSE** and persisted conversation history in **BigQuery** for multi-turn context retrieval and compliance logging.
- Built a production **RAG system** using **Pinecone** with a **hybrid retrieval architecture** combining dense embeddings (**gemini-embedding-001**, 1536-dim) and sparse vectors (**pinecone-sparse-english-v0**); implemented heading-aware chunking with breadcrumb prefixes, domain namespace isolation, and metadata-driven region filtering with Global fallback; engineered incremental re-ingestion to skip unchanged content, deployed on **Google Cloud Run** with streaming answers, indexing **100k+ enterprise vectors** across HR, IT, and RC Support domains.
- Engineered production **MCP servers** for **Claude**, enabling enterprise AI access to **Atlassian** (Jira, Confluence) and **Salesforce**; built a custom **OAuth 2.1** authentication layer (Google-backed) with per-user credential management via a **Firestore**-persisted PAT registry and **JWT Bearer** auth for Salesforce; deployed both servers to **Google Cloud Run** over **Streamable HTTP**, enabling Claude to execute live queries and mutations against enterprise systems.
- Engineered a **Document Analytics** agent using **Google Gemini 2.5** via **Vertex AI** with a **Local Execution Architecture** to process large datasets beyond context limits; built a **regex-based query sanitization layer** to intercept and validate LLM-generated **Pandas** code before execution, ensuring **100% calculation accuracy** and eliminating hallucination risk.
- Engineered a conversational **NetSuite AI Agent** using a hybrid model architecture (Gemini Flash-lite for intent routing, Flash for response generation) to enable natural language querying of procurement data; built a **PostgreSQL-backed** session management system to persist multi-turn conversation state across complex context switches.

Software Development Engineer 1 (Integrations & Automations Team)

- Automated enterprise procurement and sales workflows by integrating RingCentral with **Coupa** and **Salesforce** via an **AWS Serverless** architecture (**Lambda**, **DynamoDB**, **SQS**) in **TypeScript**; engineered real-time interactive approval cards enabling **executives** to expedite decisions, reducing approval times by **60%** and eliminating email bottlenecks.
- Developed enterprise **Ebonding** workflows and custom **REST APIs** using **Dell Boomi** for B2B partners; establishing robust integrations between **Workday**, **NetSuite**, **Salesforce**, **Jira**, **Freshservice**, and **DWH** to facilitate seamless bi-directional data synchronization.

SKILLS

Generative AI: LangGraph, LangChain, Multi-Agent Systems, Agentic AI, RAG, Pinecone, MCP, Prompt Engineering

Languages & Backend: Python (FastAPI, Flask, Pandas), TypeScript (Node.js), C++, SQL (MySQL, PostgreSQL)

AI-Assisted Frontend: React, TypeScript (Production-deployed UIs using AI coding tools)

Cloud & DevOps: GCP (Vertex AI, Cloud Run, BigQuery, Cloud Build, Firestore), AWS (Lambda, DynamoDB, SQS), Docker, Git, CI/CD

Architecture & Integration: Server-Sent Events (SSE), REST APIs, Serverless, Message Queues, Dell Boomi (iPaaS)

AWARDS & ACHIEVEMENTS

- Awarded **Ace of the Quarter** (Q1 & Q3, 2024) for exemplary performance and cross-functional collaboration.
- Mentored and guided **4 engineering interns**, resulting in successful conversion to full-time SDE roles.
- Secured All India Rank **186** in **JEE Advanced 2018**, alongside ranks **77** (Paper 2) and **585** (Paper 1) in **JEE Mains 2018**.